

Notes +

Part IV. Generative Learning Algorithms.

5.11?

• discriminative learning algorithm: learn $P(y|x)$ / mappings from the space of inputs X to the labels

• generative learning algorithm: model $p(x, y)$

(also: $p(y)$)

↑

class priors

⇒ Make use of Bayes rule: $p(y|x) = \frac{p(x|y)p(y)}{\underbrace{p(x)}_{\text{doesn't matter}}}$

$$\arg \max_y p(y|x) = \arg \max_y \frac{p(x|y)p(y)}{p(x)} = \arg \max_y p(x|y)p(y)$$

1. Gaussian discriminant analysis

Assumption: $x|y \sim N(\vec{\mu}, \vec{\Sigma})$

$y \sim \text{Bernoulli}(\phi)$

$x|y=0 \sim N(\vec{\mu}_0, \vec{\Sigma})$

$x|y=1 \sim N(\vec{\mu}_1, \vec{\Sigma})$

$$\begin{aligned} \Rightarrow \ell(\phi, \vec{\mu}_0, \vec{\mu}_1, \vec{\Sigma}) &= \log \prod_{i=1}^m P(x^{(i)}, y^{(i)}; \phi, \vec{\mu}_0, \vec{\mu}_1, \vec{\Sigma}) \\ &= \log \prod_{i=1}^m P(x^{(i)} | y^{(i)}; \vec{\mu}_0, \vec{\mu}_1, \vec{\Sigma}) P(y^{(i)}; \phi) \end{aligned}$$

$$\text{MLE: } \phi = \frac{1}{m} \sum_{i=1}^m \mathbb{1}\{y^{(i)} = 1\}$$

(personally think there's some error in the lecture note, $\vec{\mu}$ should be vector-valued)

$$\vec{\mu}_0 = \frac{\sum_{i=1}^m \mathbb{1}\{y^{(i)} = 0\} \vec{x}^{(i)}}{\sum_{i=1}^m \mathbb{1}\{y^{(i)} = 0\}}$$

$$\vec{\mu}_1 = \frac{\sum_{i=1}^m \mathbb{1}\{y^{(i)} = 1\} \vec{x}^{(i)}}{\sum_{i=1}^m \mathbb{1}\{y^{(i)} = 1\}}$$

$$\vec{\Sigma} = \frac{1}{m} \sum_{i=1}^m (\vec{x}^{(i)} - \vec{\mu}_{y^{(i)}})(\vec{x}^{(i)} - \vec{\mu}_{y^{(i)}})^T$$

⇒ We omit the process and give the result:

$$p(y=1|x; \phi, \Sigma, \mu_0, \mu_1) = \frac{1}{1 + \exp(-\theta^T x)}$$

exactly the form of logistic regression

(a discriminative algorithm)

Analysis

$p(x|y)$ multivariate gaussian $\xrightarrow{\checkmark}$ $p(y|x)$ follows a
(with shared Σ) $\xleftarrow{\times}$ logistic function

⇒ LDA makes stronger modeling assumptions $\Rightarrow$ better when data applies.

In contrast, logistic regression is more robust

1. Naive Bayes: X_i 's are discrete-valued.

text classification:

email $\rightarrow$ feature vector $\vec{x} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$

len = number of words in dict
contain
 $x_i = \begin{cases} 1 & \text{contain} \\ 0 & \text{not} \end{cases}$

Motivation: A generative model

$\Leftrightarrow$ model $\underbrace{p(x|y)}$, the length of the dictionary
 $\downarrow$ a multinomial model would have $|V| - 1$ parameters

Strong Assumption: NB

$$p(x_i | y) = p(x_i | y, x_j) \quad \forall i, j$$

(Not: $p(x_i) = p(x_i | x_j)$)

x_i & x_j are conditionally independent given y

$$p(x_1, \dots, x_n | y) = \prod_{i=1}^n p(x_i | y)$$

Parameterize: $\phi_{i|y=1} = p(x_i=1 | y=1)$
 $\phi_{i|y=0} = p(x_i=1 | y=0)$
 $\phi_y = p(y=1)$

training set: $\{x^{(i)}, y^{(i)}\} \Rightarrow L(\phi_y, \phi_{y|y=0}, \phi_{i|y=1}) = \prod_{i=1}^n p(x^{(i)}, y^{(i)})$

MLE: $\hat{\phi}_{i|y=1} = \frac{\sum_i 1\{x_i^{(i)}=1 \wedge y^{(i)}=1\}}{\sum_i 1\{y^{(i)}=1\}}$

$$\hat{\phi}_{i|y=0} = \frac{\sum_i 1\{x_i^{(i)}=1 \wedge y^{(i)}=0\}}{\sum_i 1\{y^{(i)}=0\}}$$

$$\hat{\phi}_y = \frac{\sum_i 1\{y^{(i)}=1\}}{n}$$

$$\Rightarrow \text{Prediction } P(Y=1 | x) \\ = \frac{P(x | Y=1) P(Y=1)}{P(x)} \quad \dots$$

• discretize

$\Rightarrow$ • Improvement: Laplace smoothing

Try to solve: word not in training set

$$\text{thus, MLE } \hat{\phi}_j | Y=1 = 0$$

$$\hat{\phi}_j | Y=0 = 0$$

$$\Rightarrow \text{prediction } P(Y=1 | x) = \frac{0}{2} !$$

Laplace: Consider estimate the mean of Z (r.v.)
 $\downarrow$
 take values in $\{1, 2, \dots, k\}$.

Given m independent observations: $\{Z^{(1)}, Z^{(2)}, \dots, Z^{(m)}\}$

$$\phi_j = \frac{\sum_{i=1}^m 1\{Z^{(i)} = j\}}{m}$$

$\Rightarrow$ Cause a problem: may equal to 0

$$\Rightarrow \text{LS: } \phi_j = \frac{\sum_{i=1}^m 1\{Z^{(i)} = j\} + 1}{m + k}$$

$\downarrow$ ensure $\sum \phi_j = 1$

$$\Rightarrow \text{Naive Bayes:}$$

$$\phi_j | Y=1 = \frac{\sum_{i=1}^m 1\{X_d^{(i)} = 1 \wedge Y^{(i)} = 1\} + 1}{\sum_{i=1}^m 1\{Y^{(i)} = 1\} + 2}$$

$$\phi_j | Y=0 = \frac{\sum_{i=1}^m 1\{X_d^{(i)} = 1 \wedge Y^{(i)} = 0\} + 1}{\sum_{i=1}^m 1\{Y^{(i)} = 0\} + 2}$$

• Event Models for Text Classification.

different assumptions.

① Naive Bayes : multi-variate Bernoulli event model

probability of a message. $P(y) \prod_{i=1}^n P(x_i | y)$

$\rightarrow$ through the dictionary

② multinomial event model.

$\tilde{x}_i$ denotes the identity of the i -th word in the email
take values from the dict. $\{1, \dots, |V|\}$

length of dict

$$P(y) \prod_{i=1}^n P(x_i | y)$$

$P(x_i | y)$ be interpreted as generating x_i from some multinomial distribution.